

MAT7058 Combinatorics II, Exercises 2

Due Date: 18 March 2026, Wednesday, beginning of class.
Only handwritten paper submissions in English are accepted.

- Each exercise is worth 4 points.
 - In general, you have to prove everything and only use things which were introduced in class. Ask if you are unsure.
- 1** Cars $C_1, C_2, \dots, C_n$ want to park on a one-way street with ordered parking spaces $1, 2, \dots, n$. Each car C_i , $1 \leq i \leq n$, has a preferred space a_i (here $a_i \in [n]$). The cars follow these rules:
- (i) Each car goes to its preferred parking spaces if it is empty.
 - (ii) Otherwise, it moves forward and parks in the next available space.
 - (iii) If there is no space available, then the car leaves the street and fails to park.
- The preference sequence $(a_1, a_2, \dots, a_n)$ is called a *parking function* of length n if all the cars can park. For example, $(3, 2, 1)$ is a parking function of length 3, while $(3, 2, 2)$ is not. Prove that the number of parking functions of length n is $(n+1)^{n-1}$.
- 2** Show that if $R(r-1, s)$ and $R(r, s-1)$ are both even, then $R(r, s) \leq R(r-1, s) + R(r, s-1) - 1$.
- 3** Let s_n denote the least positive integer such that in any partition of $\{1, 2, \dots, s_n\}$ into s subsets, there is a subset S in the partition with $x, y, z \in S$ such that $x+y = z$. In class we will prove that s_n exists.
- (i) Prove that $s_1 = 2$, $s_2 = 5$, $s_3 = 14$.
 - (ii) Show that $s_n \geq 3s_{n-1} - 1$.
 - (iii) Show that $s_n \geq \frac{1}{2}(3^n + 1)$.
- 4** Use the Hales-Jewett Theorem to show that there exists an integer N such that any r -coloring of $\{1, 2, \dots, N\}$ contains a monochromatic 3-term arithmetic progression $a, a+d, a+2d$ with $d \neq 0$. (Hint: Use an alphabet of size 3 and write each number in its base 3 representation.)
- 5** Show that if a graph on n vertices has e edges, then it has at least $\frac{e}{3n}(4e - n^2)$ triangles.
- 6** Let $G = (V, E)$ be a graph on n vertices which does not contain four vertices v_0, v_1, v_2, v_3 with $\{v_i, v_j\} \in E$ precisely if $i - j \equiv 1 \pmod{4}$. Show that G has at most $\frac{n}{4}(1 + \sqrt{4n-3})$ edges.